

Foundation Model-Based Zero-Shot Tissue Segmentation of Pathological Images via the Mixture of Local-to-Global Experts

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Abstract—Tissue segmentation in pathological images plays a crucial role for the diagnosis and prognosis of human cancers. However, due to the complexity of tumor micro-environment, it is difficult to annotate all tissue types especially for the categories with small tissue proportions, which limits the ability of the traditional tissue segmentation models to these tissue types with zero training samples. To address the above issues, we present a novel architecture, ZSPMLG, that relies on pathology vision-language foundation model (*i.e.*, CONCH) to learn pixel-wise classifiers for both seen and unseen tissue types based on their text descriptions. Specifically, we firstly apply large language model (LLM) to generate the descriptions for both seen and unseen tissue categories, followed by feeding them to the CONCH text encoder to acquire their corresponding prototypes that are shared by both vision and semantic space. By considering that the textual descriptions of specific tissue categories can be observed from the pathological images at different scales of magnification, our ZSPMLG consists of Mixture of Local Experts (MoLE) and Mixture of Global Experts (MoGE) modules, where MoLE performs the specialized decoding that can map individual scale patch-level representation to dense pixel-level representation, while MoGE aims at fusing the multi-scale representations together. Finally, a convolutional layer is designed to map the pixel-level representation to the category prototype for tissue segmentation on both seen and unseen categories. We evaluate our method on three datasets and the experimental results demonstrate the superiority of our method on both seen and unseen tissue categories.

Index Terms—Zero-shot tissue segmentation, foundation model, histopathological images.

I. INTRODUCTION

TISSUE segmentation is a fundamental task in computational pathology that focuses on the delineation and classification of distinct tissue regions (*i.e.*, tumors, stroma, and lymphocytic infiltrates) within histological samples [1],

[2], [3], [4], [5]. This task is essential for accurate cancer diagnosis, prognosis assessment, and treatment planning as it can facilitate the quantitative analysis of tissue architecture and tumor micro-environments [6], [7], [8], [9], [10].

Nowadays, the rapid advancement of deep learning has revolutionized the field of computer vision [11], [12], [13], [14], inspiring significant progress in the development of automated tissue segmentation techniques. For instance, Liu et al. [15] have presented a multi-stained multi-tissue segmentation algorithm on kidney histology images via generative self-supervised meta-learning framework. Van Rijthoven et al. [16] introduce the HookNet that can combine context and details via encoder-decoder branches for pathology tissue segmentation. However, most of the existing tissue segmentation models rely heavily on supervised learning paradigms, which requires extensive and labor-intensive annotations provided by expert pathologists. In fact, due to the complexity of tumor micro-environment, it is difficult to annotate all tissue categories in pathological images. For instance, while most of the existing tissue segmentation models are designed to distinguish between tumor and stromal regions in pathology images [16], gaining deeper insights into tumor progression and metastasis requires analysis of the spatial distribution of lymphocytes that are related to the immune response to human cancers [17]. However, lymphocytes are often small and difficult to annotate as they are sparsely distributed in the pathological images, which underscores the pressing need to develop the zero-shot tissue segmentation model that can identify unseen and scarcely annotated tissue types without requiring task-specific labeled data [18].

Recently, large-scale vision-language pre-training models like CLIP [19] have shown impressive transferability in recognizing novel categories, leading to their increased adoption for tackling the challenging zero-shot segmentation (ZSS) task [20], [21]. Generally, they utilize the text prompts to represent the unseen categories and leverage the CLIP's alignment ability between visual and textual information to learn visual representation of unseen categories for zero-shot segmentation. For instance, ZegFormer [22] takes full advantage of the pre-trained vision-language model by decoupling the zero-shot semantic segmentation into class-agnostic segmentation and segment-level zero-shot classification tasks. Zhou et al. [20] pursue a simpler-and-efficient one-stage ZSS model that directly extends CLIP's zero-shot prediction capability from

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image to pixel level by adding a lightweight transformer. Other studies like MaskCLIP+ apply CLIP to generate pseudo annotations on novel classes via self-training and have further broadened segmentation research to more complex settings [23], [24], [25], [26], [27], such as interactive 3D segmentation [26] and class-incremental semantic image segmentation [27].

Although much progress has been achieved, these vision-language based ZSS approaches are primarily designed on natural images and overlook the fact that the text descriptions of different tissue categories in the pathological images can be observed from the images at different scales of magnification. For example, the tissue of *lymphocytes* could be described as “with dark and round nuclei” and “scattered throughout the tissue”, where the former text descriptions could be revealed by the tissue images at high magnification level, while the latter descriptions are visible at low magnification level. In addition, most of the existing ZSS models rely on the unified decoder, which has difficulty in obtaining diverse tissue structures with distinct characteristics. Moreover, due to the lack of visual exemplars of unseen category tissues and the complex structure of different tissue types, it is still challenging to establish accurate semantic-visual correspondences for the representation of unseen tissue categories during the inference stage.

Based on the above consideration, we present a novel architecture, ZSPMLG, that relies on pathology vision-language foundation model (*i.e.*, CONCH [28]) to learn pixel-wise classifiers for both seen and unseen tissue types according to their text descriptions generated by large language model (LLM). ZSPMLG applies the text encoder of CONCH to generate the prototypes for both seen and unseen tissue categories via large language models. The image branch of ZSPMLG consists of Mixture of Local Experts (MoLE) and Mixture of Global Experts (MoGE) modules, where MoLE considers the complex and diverse structure of pathological images that can map individual scale patch-level representations to dense pixel-level representation. Meanwhile, the MoGE module aims at effectively fusing the multi-scale representations through a gating mechanism. Finally, we map the pixel-level representation to the category prototypes for tissue segmentation on both seen and unseen categories. We evaluate our method on three pathology image datasets for tissue segmentation with different number of unseen categories, and the experimental results clearly show the advantage of our method in comparison with the existing studies. The main contributions of this study are listed as follows:

1) We combine large language models (LLMs) with pathology vision-language foundation model (CONCH) to generate semantic prototypes for both seen and unseen tissue categories, enabling effective pixel-level representations for tissue segmentation.

2) By considering that the textual descriptions of different tissue categories could be observed from different scale of magnifications, our ZSPMLG integrates a MoLE module for decoding scale-specific patch-level features to pixel-level representation and a MoGE module for the fusion of multi-scale representations.

3) We have developed a novel patch-to-pixel decoder architecture that enables fine-grained alignment between visual features and semantic prototypes to capture diverse tissue structures at the pixel-level for tissue segmentation.

4) Unlike the general VLM-based segmentation methods that rely on simple class names, our framework fully considers the inherent complexity of histopathological images and leverages LLM to generate comprehensive, multi-granularity semantic descriptions for each tissue type.

II. RELATED WORKS

A. Tissue Segmentation in Pathological Images

Tissue segmentation is a fundamental task for the analysis of histopathological images that aims at segmenting regions with certain characteristics and structures (*i.e.*, gland, tumor-infiltrating lymphocytes, etc.) [29], [30]. In recent years, deep learning has demonstrated remarkable success in various computer vision and image processing tasks [11]. Motivated by its superior performance, an increasing number of studies have explored the application of deep learning techniques to the segmentation of diverse tissue types in whole slide pathological images (WSIs). For instance, Liu et al. [15] have presented a generative self-supervised meta-learning framework for multi-stained multi-tissue segmentation in renal biopsy whole slide images. Lu et al. utilize deep learning to perform automatic segmentation and quantification of epithelial and stromal tissues for breast cancer [31]. Building upon U-Net, a deep contour-aware network is developed for the gland tissue segmentation on pathological images [32]. Saltz et al. present a minimal information loss dilated network to acquire the spatial distributions of tumor-infiltrating lymphocytes on H&E images from 13 TCGA (The Cancer Genome Atlas) tumor types [33]. By considering that the multi-scale decoder can capture both global context and local details for tissue segmentation, the studies in [34] and [35] introduce a series of multi-decoder fully convolutional neural networks with deep fusion for tissue segmentation in histopathology images. Hatamizadeh et al. [36] propose a hierarchical transformer-based segmentation framework that connects multi-resolution encoder representations to a CNN decoder. However, these multi-scale decoders rely on a static fusion mechanism, leaving them incapable of dynamically selecting scale-specific experts based on the input context. To address it, Hu et al. [37] present a dynamic knowledge-guided multi-granularity multi-scale fusion model for histological segmentation. Gu et al. [38] propose an adaptive spatial and frequency experts network for multi-scale medical image fusion. More recently, by considering that the existing tissue segmentation methods require huge annotation efforts from experts, several semi-supervised or weakly-supervised tissue segmentation algorithms are developed. Specifically, Guan et al. [39] propose a global-local consistent semi-supervised tissue segmentation algorithm on pathological images. By implementing a self-supervised task that focuses on uncovering non-discriminative parts of objects, Kang et al. [40] develop an efficient weakly-supervised tissue segmentation method with image-level labels to reduce the annotation cost. However, all these studies assume that each

target tissue categories should be annotated in the training set. In fact, it is difficult to annotate all tissue categories in pathological images especially for the small proportion tissue types that are hard for annotation.

B. Zero-Shot Segmentation

Zero-shot segmentation (ZSS) is a challenging computer vision task that aims to segment objects or regions belonging to previously unseen categories [41], [42], [43], [44]. Recently, with the fast development of foundation models, the existing ZSS studies can be categorized into two streams *i.e.*, Segment Anything (SAM) and vision-language based models. On one hand, since SAM is trained by over 1 billion masks on 11 million images that makes it attractive for zero-shot segmentation, Kirillov et al. [45] have presented a novel approach for minimally-guided zero-shot segmentation using SAM, Zhang et al. [46] propose a no-external prompt SAM-based method for zero-shot texture anomaly detection, called Image Internal Information Mining SAM (IIIM-SAM). However, most of the existing SAM based methods ignore the fact that the semantic information is also helpful for the ZSS task. Motivated by the significant advancements in vision-language models (VLMs), recent studies have sought to extend their strong zero-shot generalization capabilities from image-level classification to the fine-grained task of pixel-level segmentation. For instance, Koleilat et al. have presented a novel framework, called MedCLIP-SAM that combines CLIP and SAM models to generate segmentation of clinical scans using text prompts in both zero-shot and weakly supervised settings [47]. Ding et al. design a simple but effective ZSS model that uses a transformer decoder to generate segment-level embeddings for grouping and zero-shot classification [22]. Building on the idea of special tokens in language modeling representing sentence-level embeddings, the study in [21] introduces a trusty token that enables distinguishing novel classes from known ones for ZSS task. As to the zero-shot segmentation on the medical images, the studies in [48] and [49] have presented an in-depth exploration of integrating SAM and CLIP into a unified framework for zero-shot medical image segmentation. Hamrani and Godavarty [50] introduce an unsupervised learning system that uses VLM to segment biomedical images without needing any prior labels. Moreover, recent pathology foundation models have shown substantially advancement for zero-shot pathological image analysis, the studies in [28], [51], and [52] leverage pathology image-text pairs to build pathology foundation models for multi-modal representations. Built upon the pathology foundation model, various downstream tasks such as classification [53], survival prediction [54] and molecular subtype inference [55] can be readily developed with minimal task-specific adaptation. However, most of these pathology foundation models have not been applied to zero-shot pathological image segmentation task. To the best of our knowledge, only Tang et al. [56] attempt to address zero-shot segmentation of unseen pathological tissue types by leveraging low-resolution whole-slide images (WSIs), yet their approach largely overlooks the multi-scale nature of pathological features, where the textual descriptions of various tissue categories correspond

to visual features that manifest at different magnification scales. In summary, the existing foundation-model-based zero-shot segmentation methods either lack semantic understanding (SAM-based) [48] or remain ill-suited for pathology due to their disregard for multi-scale features (VLM-based) [49]. To bridge this gap, we propose a novel framework that jointly leverages the strengths of pathology foundation model while explicitly incorporating multi-scale visual-textual alignment tailored to the tissue segmentation in pathology images.

III. METHOD

A. Overview of the Proposed Method

Fig. 1 shows the flowchart of the proposed ZSPMLG framework, which is built upon the pathology-language model CONCH [28]. For the text branch, category prototypes for both seen and unseen tissue types are generated using the CONCH text encoder. In the image branch, we introduce two key modules: MoLE, which projects patch-level features to dense pixel-level representations at individual scales, and MoGE that can integrate multi-scale representations. Finally, pixel-level features are aligned with the corresponding category prototypes to enable segmentation across both seen and unseen tissue categories.

B. Problem Formulation

For ease of representation, we denote the set of seen (unseen) categories as $C^s(C^u)$ and $C^s \cap C^u = \emptyset$. In the zero-shot segmentation task, the training set only contains pixel-wise annotations of C^s , while the trained model is supposed to segment objects with categories $C^s \cup C^u$ at test time. In this paper, the bridge between the seen and unseen tissue categories is the prototypes extracted from the word embeddings of their category-level descriptions. We follow the transductive settings, which means that we can apply the text descriptions of the unseen categories for model training but their corresponding pixel-level annotations are still unknown to us.

C. Text Branch for ZSPMLG

In the text branch of our ZSPMLG, we hope to learn category prototypes from their corresponding text descriptions. These prototypes are intended to bridge the semantic gap between seen and unseen tissue categories, enabling generalization beyond the training classes. Here, the text description of different tissue categories can be obtained via GPT-4o [57], and we generate the text descriptions for different tissue categories by considering their appearance with the following prompts: “Describe {} in the histopathology image in detail, including features such as color, shape and arrangement.” where the placeholder is replaced by the tissue name, such as “tumor”, “stroma”, etc. Given the description set $\hat{D} = \{\hat{d}_1, \hat{d}_2, \dots, \hat{d}_{n_c}\}$, where n_c is the number of categories ($n_c = |C^s| + |C^u|$), we utilize a frozen language encoder denoted as $f_{\text{text}}(\cdot)$ from the pathology-language model CONCH [28] to effectively map the descriptions of each tissue category into

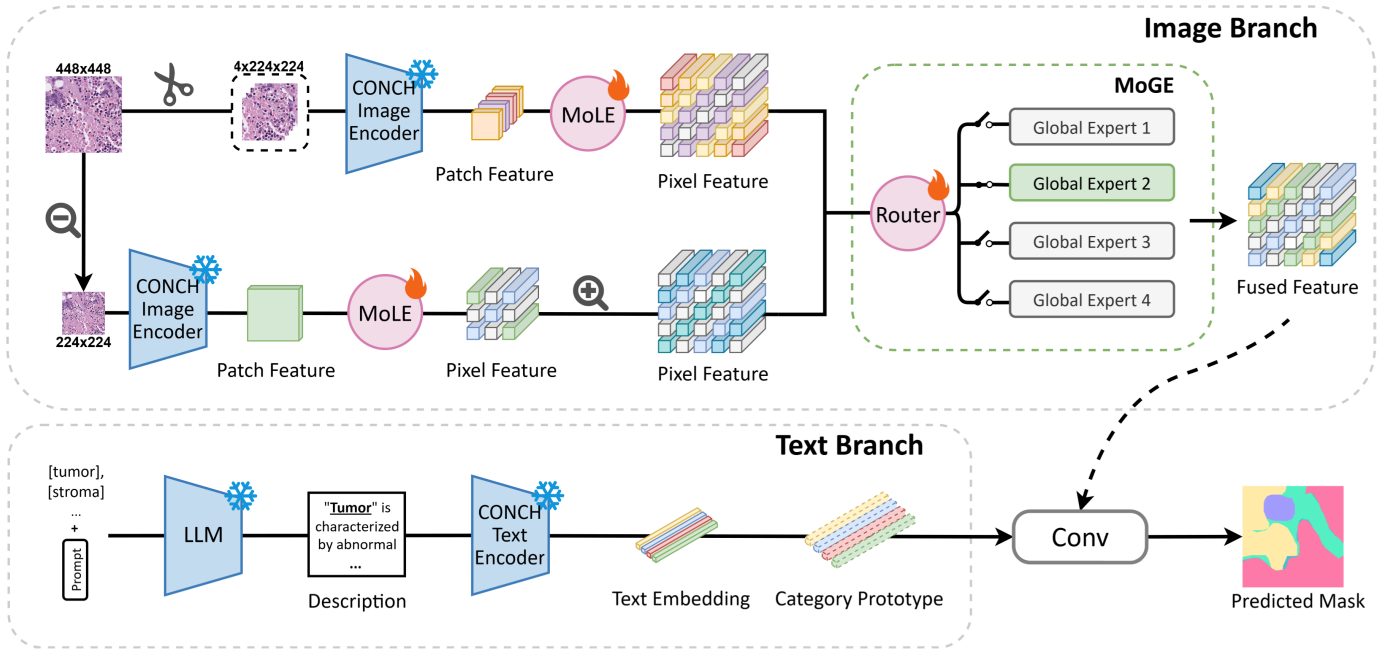


Fig. 1. The flowchart of the proposed ZSPMLG model. For the text branch, we generate the prototypes for both seen and unseen tissue categories via the CONCH text encoder. As to the image branch, we design the MoLE (Mixture of Local Experts) and MoGE (Mixture of Global Experts) modules that can map the patch-level representations to the dense pixel-level representations at individual scale and fuse multi-scale representations, respectively. Finally, we map the pixel-level representation to the category prototype for the segmentation of both seen and unseen categories.

their corresponding high-dimensional semantic embeddings $\hat{T} \in \mathbb{R}^{512 \times n_c}$:

$$\hat{T} = [\hat{t}_1, \hat{t}_2, \dots, \hat{t}_{n_c}] \in \mathbb{R}^{512 \times n_c} \quad (1)$$

where $\hat{t}_i = f_{\text{text}}(\hat{d}_i)$. Then, for the purpose of aligning the semantic embeddings with the visual representations (introduced in the following Section III-D), we design a text-to-prototype projector *i.e.*, $f_{\text{proj}}(\cdot)$ consisting of two linear layers to transform the semantic representation of each tissue category $\hat{t}_i$ to its corresponding prototypes that are shared by both image and text branches.

$$f_{\text{proj}}(\hat{t}_i) = W_2 \cdot \text{ReLU}(W_1 \cdot \hat{t}_i + b_1) + b_2 \quad (2)$$

where $W_1 \in \mathbb{R}^{256 \times 512}$, $W_2 \in \mathbb{R}^{64 \times 256}$, $b_1 \in \mathbb{R}^{256}$, $b_2 \in \mathbb{R}^{64}$ are learnable parameters, and $\sigma(\cdot)$ is a nonlinear *ReLU* activation function. Finally, the prototype matrix P for different tissue categories can be represented as:

$$P = f_{\text{proj}}(\hat{T}) = [p_1, p_2, \dots, p_{n_c}] \in \mathbb{R}^{64 \times n_c} \quad (3)$$

D. Image Branch for ZSPMLG

For the image branch, we consider that the text descriptions of different tissue categories can be observed from the pathological images at different scales of magnification, and thus we propose a dynamic fusion framework based on the mixture of local-to-global experts to combine the pixel-level representations extracted from different scales. Specifically, as shown in Fig. 1, given the m high-resolution cropped regions with the size of 448×448 from the pathological images $R = [R_1, R_2, \dots, R_m]$, we firstly equally split each region into 4 non-overlapping patches with the size of 224×224 . After

feeding all the $4m$ high-magnification patches to the CONCH image encoder for the acquirement of their embeddings:

$$I^h = [I_1^h, I_2^h, \dots, I_{4m}^h] \in \mathbb{R}^{768 \times 4m} \quad (4)$$

We reshape I^h into its feature map $FI^h \in \mathbb{R}^{768 \times M^h \times M^h}$, where $M^h = 2\sqrt{m}$. As to the pathway of low magnification level, we firstly downsample each region in R to patches with the size 224×224 . Then, we follow the step in the pathway of high magnification level to derive its corresponding feature map $FI^l \in \mathbb{R}^{768 \times M^l \times M^l}$, where $M^l = \sqrt{m}$. In what follows, we will introduce MoLE module for pixel-level representation, and MoGE module for the fusion of multi-scale image features for tissue segmentation.

1) *Mixture of Local Experts Module*: For the purpose of transforming the patch-level features to the pixel-level representation, we design a patch-to-pixel decoder based on MoLE module for each scale of magnification. Specifically, MoLE firstly incorporates a local gating network G_{local} and a set of 4 local expert networks $[C_1, \dots, C_4]$. The structure of each expert network is a Conv(1 × 1)-BN-ReLU block. The gate network G_{local} takes the feature map $FI^k(k \in \{h, l\})$ of each individual magnification as input and produces the weight of each expert $G_{\text{local}}(FI^k)$ as follows:

$$G_{\text{local}}(FI^k) = [(z_{\text{local}}^k)^1, \dots, (z_{\text{local}}^k)^4] = \sigma(\mathcal{W} \cdot FI^k) \in \mathbb{R}^4 \quad (5)$$

where $\mathcal{W}$ denotes the trainable weight matrix and $\sigma(\cdot)$ represents softmax operation. Each expert network C_i processes FI^k independently and then the result of mixture of experts can be denoted as:

$$y_{\text{local}}^k = \sum_{i=1}^4 (z_{\text{local}}^k)^i \cdot C_i(FI^k) \quad (6)$$

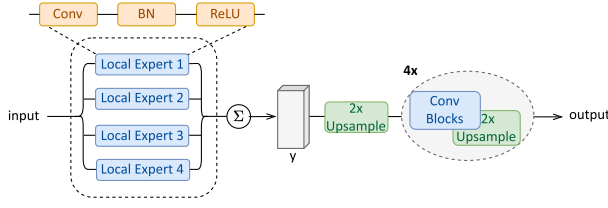


Fig. 2. The structure of the patch-to-pixel decoder based on MoLE.

Here, $C_i(FI^k)$ indicates the output of the i -th local expert network on FI^k . Next, 4 conv-upsampling blocks (shown in Fig. 2) are designed for mapping y_{local}^k to the pixel-level representations $FP^k (k \in \{l, h\})$ on each magnification level. Each block progressively increases the spatial resolution until the final output dimensions are obtained. Since the size of $FP^l \in \mathbb{R}^{64 \times 224 \times 224}$ is sub-scale to $FP^h \in \mathbb{R}^{64 \times 448 \times 448}$, we further upsample FP^l to ensure the dimension consistency between them for the following multi-scale fusion step. It is worth noting that although the experts in MoLE module share the same network architecture, prior studies [58] have shown that the gating mechanism enables them to specialize on different parts of the input data. We will exhibit patches with diverse tissue structures that different experts focus on according to the weight of each expert $G_{local}(FI^k)$ in Section IV-E.

2) *Mixture of Global Experts Module*: Next, for the purpose of effectively combining the pixel-level representations derived from different scales, MoGE module is designed. Specifically, MoGE includes a global gating network G_{global} and a set of 4 global expert networks $[E_1, E_2, E_3, E_4]$, where each expert network represents a specific fusion strategy for the combination of pixel-level representations derived from high magnification (*i.e.*, FP^h) or low magnification (*i.e.*, FP^l) pathological images as follows:

$$\begin{aligned} E_1 &\triangleq \mathcal{DS}(\mathcal{SA}(\text{concat}(FP^h, FP^l))) \\ E_2 &\triangleq \mathcal{SA}(FP^h + FP^l) \\ E_3 &\triangleq \mathcal{CA}(FP^h, FP^l) \\ E_4 &\triangleq \mathcal{DS}(\mathcal{SA}(\text{concat}(\mathcal{SA}(FP^h), \mathcal{SA}(FP^l)))) \end{aligned} \quad (7)$$

Where $\mathcal{DS}$ represents 2×2 downsampling operator, $\mathcal{SA}$ and $\mathcal{CA}$ are self-attention and cross-attention operators, respectively. As indicated in Eq. (7) and Fig. 3, each expert E_i corresponds to a specific fusion style. E_1 corresponds to the fusion strategy of early concatenation, where the token sequences from different magnification levels are concatenated and jointly processed by self-attention. E_2 implements early summation, in which the feature maps FP^h and FP^l are directly summed and then passed through a self-attention module. E_3 directly adopts a cross-attention strategy to capture the association among multi-scale features. Moreover, E_4 firstly applies the self-attention on individual scale representation, and then concatenates them together for the derivation of the final outputs by another self-attention operator. For E_1 and E_4 ,

we apply $\mathcal{DS}$ to ensure that the output of all experts are with the same dimensionality.

To select the most suitable expert for multi-scale representation fusion, we employ a lightweight global gating network G_{global} denoted as follows, which takes the concatenated feature map $[FP^h, FP^l]$ as input. The entire process can be formulated as:

$$G_{global}(FP^l, FP^h) = \mathcal{P}(\text{concat}([FP^h, FP^l])) \in \mathbb{R}^4 \quad (8)$$

where $\mathcal{P}$ refers to the network including two 3×3 convolutional layers with output channels equal to 64 and 32 respectively, each followed by ReLU activation; an adaptive average pooling layer that reduces the spatial dimension to 1×1 ; a fully connected layer projecting the pooled features to a 4-dimensional vector indicating the weights of different experts. We choose the expert with the largest weight to derive the fused pixel-level representations denoted as $\mathcal{I}_{\mathcal{F}} \in \mathbb{R}^{64 \times 448 \times 448}$.

The entire MoGE module is trained end-to-end along with the entire image branch. Although only one expert is active per forward pass, gradients are back-propagated through both the selected expert and the gating network. This enables the gating network to learn effective routing policies, while each expert learns to specialize on different input patterns.

E. Prototype-Guided Segmentation

Given the fused pixel-wise representations $\mathcal{I}_{\mathcal{F}}$ and category prototypes P with shape $64 \times n_c$ (see section III-C), we perform tissue segmentation for both seen and unseen categories by treating P as kernels followed by convolving it with $\mathcal{I}_{\mathcal{F}}$ to produce the segmentation logits, which can be formulated as:

$$S = P * \mathcal{I}_{\mathcal{F}} \in \mathbb{R}^{n_c \times H \times W} \quad (9)$$

Here $*$ is the convolution operation. This formulation enables direct alignment between visual features and semantic prototypes. The Focal Loss $\mathcal{L}_{\text{focal}}$ [59] is used to mitigate class imbalance by down-weighting well-classified pixels, while the Dice Loss $\mathcal{L}_{\text{dice}}$ [60] is employed to encourage spatial overlap between predicted and ground truth masks. Then, the final segmentation loss of our method $\mathcal{L}_{\text{seg}}$ is the combination of Focal and Dice losses denoted as:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{dice}}(M, S) + \mathcal{L}_{\text{focal}}(M, S) \quad (10)$$

where M refers to the ground truth mask of the input images.

IV. EXPERIMENTS

A. Dataset

To validate the effectiveness of our method for tissue segmentation, we conduct experiments on the three datasets (*i.e.*, TNBC, HPBC and CATCH) for zero-shot tissue segmentation on pathological images. For each of these three datasets, the annotations for all tissue categories are available. To meet the learning paradigm for zero-shot segmentation, we divide all tissue types into seen and unseen categories. We set the tissue types like tumor or stromal as seen categories since they are primary TME components in pathology tissues. For

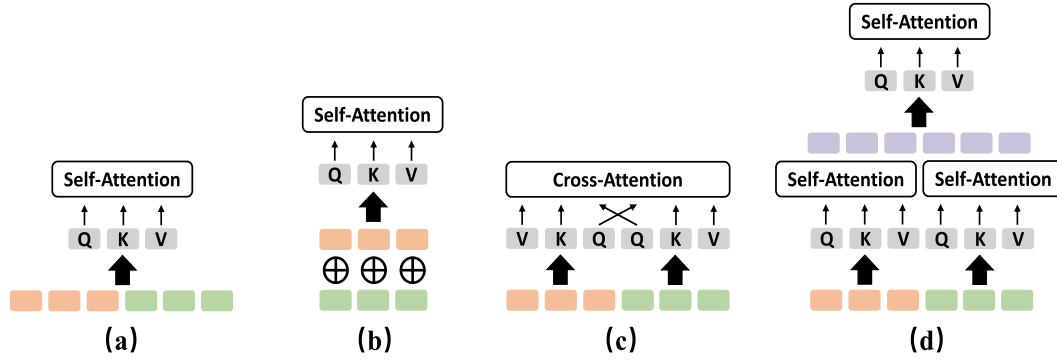


Fig. 3. Illustrations of the 4 global expert networks (*i.e.*, E_1, E_2, E_3, E_4) corresponding to 4 different ways for the combination of pixel-level representations derived from high-resolution and low-resolution pathological images.

TABLE I

COMPARISON OF ZSPMLG WITH THE EXISTING ZERO-SHOT SEGMENTATION METHODS ON TNBC DATASET. w2v: WORD2VEC. mIoU(S): mIoU ON SEEN CATEGORIES. mIoU(U): mIoU ON UNSEEN CATEGORIES. hIoU: HARMONIC mIoU. THE SYMBOL “*” INDICATES THAT OUR METHOD IS SIGNIFICANTLY SUPERIOR ($p < 0.05$) TO THE CORRESPONDING COMPARING METHOD VIA PAIRWISE T-TEST

Method	Text Encoder	TNBC											
		1 unseen				2 unseen				3 unseen			
		pAcc	mIoU(S)	mIoU(U)	hIoU	pAcc	mIoU(S)	mIoU(U)	hIoU	pAcc	mIoU(S)	mIoU(U)	hIoU
SPNet [61]	w2v	59.3(± 2.3)*	34.1(± 1.7)*	31.0(± 1.4)*	32.5(± 1.5)*	60.5(± 2.1)*	41.4(± 1.8)*	24.2(± 0.7)*	30.5(± 1.0)*	62.4(± 2.4)*	41.7 (± 0.6)*	25.4 (± 1.1)*	31.6(± 0.8)*
ZS5 [41]	w2v	60.5(± 1.2)*	36.3(± 1.3)*	33.9(± 0.8)*	35.1(± 1.0)*	59.3(± 0.6)*	40.2(± 1.1)*	29.6(± 1.2)*	34.1(± 1.1)*	62.4(± 2.2)*	40.1(± 0.9)*	29.5(± 2.6)*	34.0(± 1.3)*
CaGNet [62]	w2v	67.8(± 1.5)*	41.7(± 2.7)*	39.4(± 2.3)*	40.5(± 2.5)*	66.6(± 0.8)*	47.2(± 1.5)*	29.4(± 0.4)*	36.2(± 0.6)*	67.5(± 1.0)*	48.1(± 0.7)*	29.9(± 1.3)*	36.9(± 0.9)*
SaLIP [49]	CONCH	70.1(± 0.8)*	42.4(± 0.3)*	41.9(± 0.3)*	42.1(± 0.3)*	68.6(± 1.4)*	51.2(± 1.1)*	25.7(± 0.4)*	34.2(± 0.6)*	69.7(± 2.4)*	53.1(± 1.1)*	26.3(± 0.9)*	35.2(± 1.0)*
MaskCLIP+ [23]	CONCH	74.7(± 1.2)*	53.7(± 0.6)*	49.9(± 1.4)*	51.7(± 0.8)*	72.8(± 0.6)*	55.5(± 1.3)*	31.7(± 0.8)*	40.4(± 1.0)*	72.4(± 1.6)*	54.6(± 1.4)*	33.7(± 1.3)*	41.7(± 1.3)*
ZegCLIP [20]	CONCH	74.9(± 1.6)*	51.3(± 1.4)*	50.3(± 1.0)*	50.8(± 1.2)*	73.4(± 0.6)*	55.2(± 0.7)*	31.8(± 1.5)*	40.4(± 1.0)*	72.1(± 0.9)*	55.3(± 1.7)*	33.4(± 1.9)*	41.6(± 1.8)*
TagCLIP [21]	CONCH	74.3(± 1.6)*	53.4(± 2.5)*	51.2(± 0.7)*	52.3(± 1.1)*	73.5(± 1.6)*	56.7(± 1.3)*	32.7(± 0.6)*	41.5(± 0.8)*	72.6(± 1.4)*	55.6(± 1.5)*	33.0(± 1.1)*	41.4(± 1.3)*
ZSPMLG(ours)	CONCH	76.7(± 1.5)	57.1(± 0.9)	54.1(± 1.3)	55.6(± 1.1)	75.3(± 1.4)	58.2(± 1.6)	36.2(± 1.4)	44.6(± 1.5)	74.6(± 2.1)	57.0(± 0.7)	35.2(± 1.6)	43.5(± 1.0)

the categories with small tissue proportions that are not widely noticed and difficult for annotation, we set them as unseen categories and hope to leverage vision-language model to realize accurate tissue segmentation. More specifically, the **TNBC** (Triple-Negative Breast Cancer) dataset [63] contains 1,393 pathological image patches of size 448×448 and are annotated with six different tissue types: tumor (tum, 45%), stroma (str, 34%), lymphocytic infiltrate (lym, 10%), necrosis (nec, 7%), plasma cells (pla, 2%) and fat (2%). In our experiments, the settings of unseen classes are denoted as **1 unseen**:[fat], **2 unseen**:[pla, fat] and **3 unseen**:[nec, pla, fat]. **HPBC** is an in-house pathological image dataset collected from Nanjing Drum Tower Hospital for Her2-positive breast cancer. HPBC contains 4,956 patches of size 448×448 with 4 categories: background(bg, 25%), stroma (str, 56%), lymphocytic infiltrate (lym, 4%), and tumor (tum, 16%). We set [lym] as the **unseen class**. Finally, **CATCH** [64] is a publicly available lung cancer dataset for tissue segmentation that contains 2,665 image patches of size 448×448 with 5 categories: background (10%), subcutis (sub, 30%), peripheral nerve sheath tumor (pnst, 49%), squamous cell carcinoma (scc, 6%), and trichoblastoma (tri, 5%). In our experiments, the settings of unseen classes are defined as **1 unseen**:[tri], **2 unseen**:[scc, tri].

B. Experimental Settings and Evaluation Metrics

For each of the three datasets, we randomly select 80% images as the training dataset and the remaining 20% images

as the test dataset. In the training phase, we set the batch size as 32 with the learning rate of 0.05. In the text branch of ZSPMLG, we use GPT-4o [57] as the LLM to generate descriptions of different tissue categories (Introduced in Section III-C). As to the image branch, CONCH [28] is applied to extract patch-level representations. Moreover, we adopt a self-training strategy introduced in [20] and [21] to facilitate model training, and the number of training epochs is set as 200. To evaluate the performance of our method, two metrics (*i.e.*, pAcc and mIoU) are adopted. Specifically, pAcc measures the pixel-level classification results, and mIoU indicates the class-wise intersection over Union. For mIoU, we separately calculate it on seen and unseen categories, denoted as mIoU(S) and mIoU(U), respectively. Additionally, we also adopt the index of harmonic mean IoU(hIoU) [65] that can be derived by the combination of mIoU(S) and mIoU(U).

C. Comparison With the Existing Studies for Zero-Shot Tissue Segmentation

To demonstrate the effectiveness of the proposed ZSPMLG, we firstly compare it with the following seven state-of-the-art (SoTA) methods for zero-shot tissue segmentation in Table I and Table II. For the competitors based on the vision-language models (*e.g.*, CLIP), we replace it with CONCH for fair comparison. 1) SPNet [61]: An end-to-end semantic segmentation model which maps each image pixel to a class-level semantic embedding. 2) ZS5 [41]: A zero-shot semantic segmentation method that incorporates a graph-context encoding to leverage

TABLE II

COMPARISON OF ZSPMLG WITH THE EXISTING ZERO-SHOT TISSUE SEGMENTATION METHODS ON HPBC AND CATCH DATASETS. W2V: WORD2VEC. MIOU(S): MIOU ON SEEN CATEGORIES. MIOU(U): MIOU ON UNSEEN CATEGORIES. hIOU: HARMONIC MIOU. THE SYMBOL “*” INDICATES THAT OUR METHOD IS SIGNIFICANTLY SUPERIOR ($p < 0.05$) TO THE CORRESPONDING COMPARING METHOD VIA PAIRWISE T-TEST

Method	Text Encoder	HPBC				CATCH							
		1 unseen				1 unseen				2 unseen			
		pAcc	mIoU(S)	mIoU(U)	hIoU	pAcc	mIoU(S)	mIoU(U)	hIoU	pAcc	mIoU(S)	mIoU(U)	hIoU
SPNet [61]	w2v	66.5(±2.4)*	58.6(±1.2)*	22.0(±1.7)*	32.0(±1.4)*	60.7(±0.8)*	33.4(±1.0)*	31.7(±2.5)*	32.5(±1.4)*	59.6(±1.6)*	40.9(±1.1)*	29.5(±0.7)*	34.3(±0.9)*
ZS5 [41]	w2v	68.3(±0.8)*	56.4(±0.4)*	24.7(±0.9)*	34.4(±0.6)*	62.4(±2.1)*	37.2(±0.6)*	31.5(±1.1)*	34.1(±0.8)*	61.7(±0.8)*	39.3(±2.8)*	32.4(±1.7)*	35.5(±2.1)*
CaGNet [62]	w2v	71.1(±0.9)*	59.2(±1.5)*	30.3(±1.0)*	40.1(±1.2)*	68.9(±0.7)*	43.4(±0.7)*	38.7(±0.8)*	40.9(±0.7)*	67.4(±2.3)*	45.4(±1.9)*	33.0(±1.0)*	38.2(±1.3)*
SaLiP [49]	CONCH	70.2(±1.7)*	60.5(±1.6)*	27.9(±1.4)*	38.2(±1.5)*	70.4(±0.8)*	42.7(±0.9)*	39.9(±1.1)*	41.3(±1.0)*	69.4(±0.8)*	52.7(±2.5)*	33.4(±0.9)*	40.9(±1.3)*
MaskCLIP+ [23]	CONCH	73.5(±2.5)*	61.8(±1.6)*	33.4(±0.7)*	43.4(±1.0)*	75.8(±1.8)*	53.6(±1.3)*	50.5(±1.1)*	52.0(±1.2)*	73.4(±0.8)*	57.0(±0.6)*	37.6(±0.6)*	45.3(±0.6)*
ZegCLIP [20]	CONCH	73.1(±1.5)*	61.9(±1.7)*	35.4(±0.8)*	45.0(±1.1)*	73.6(±2.0)*	52.1(±1.2)*	50.9(±1.3)*	51.5(±1.2)*	72.4(±1.4)*	54.4(±0.9)*	38.4(±0.8)*	45.0(±0.8)*
TagCLIP [21]	CONCH	74.9(±0.9)*	62.5(±0.6)*	36.7(±1.0)*	46.2(±0.8)*	75.2(±1.8)*	55.1(±0.7)*	51.3(±1.0)*	53.1(±0.8)*	72.6(±0.7)*	56.4(±1.2)*	38.7(±1.7)*	45.9(±1.4)*
ZSPMLG(ours)	CONCH	75.9(±0.5)	64.0(±1.3)	40.7(±1.2)	49.8(±1.2)	77.1(±1.4)	56.9(±1.7)	53.5(±1.0)	55.1(±1.3)	73.9(±1.8)	58.1(±0.5)	41.7(±2.0)	48.6(±0.8)

spatial relationships among different categories. 3) CaGNet [62]: A zero-shot segmentation model that applies the pixel-level neighborhood information for unseen class segmentation. 4) SaLiP [49]: Leverages SAM to generate multiple candidate masks and uses CLIP with descriptive text to select the region of interest for zero-shot tissue segmentation. 5) MaskCLIP+ [23]: Leverages CLIP to provide pseudo labels for unlabeled pixels, which can be applied to more segmentation-tailored architectures for zero-shot segmentation. 6) ZegCLIP [20]: Adapts CLIP for zero-shot semantic segmentation through a one-stage pipeline that computes pixel-level class similarities using text embedding. 7) TagCLIP [21]: A novel trusty-aware guided CLIP for zero-shot tissue segmentation.

From the experimental results shown in Table I and Table II, we can derive the following two observations. Firstly, the studies relying on the foundation vision-language models (*i.e.*, SaLiP [49], MaskCLIP+ [23], ZegCLIP [20] and TagCLIP [21]) perform better than their counterparts (*i.e.*, SPNet [61], ZS5 [41], CaGNet [62]), showing the advantage of adopting foundation vision-language models on ZSS task. Secondly, our proposed method can achieve the best performance for tissue segmentation, particularly on the unseen classes. The reason lies in the following two aspects. Firstly, instead of simply relying on the class names, we use the text descriptions derived from GPT-4o [57] to generate the prototypes that can more accurately reflect the tissue category information for the zero-shot segmentation task. More discussions can be found in Section IV-F. Secondly, our method combines the multi-scale pathological image features to generate pixel-level representations that can more effectively reflect the category information derived from the text branch.

D. Comparisons of the Visualization Results

We also show the visualization results of different methods in Fig. 4. It is evident that in comparison with the SoTA vision-language model (TagCLIP [21]), our method can more effectively detect unseen tissues as well as achieve better segmentation performance on both seen and unseen tissue categories. For instance, as to the CATCH data under the settings of 2 unseen categories (last row in Fig. 4), our method can not only perform more precise segmentation for the known categories (*e.g.*, pnst, sub), but also accurately recognize

unknown tissue categories (*i.e.*, scc, tri) that do not appear in the training dataset. Moreover, our method is also superior to SAM [45] and MedCLIP-SAM [47] in distinguishing different tissue types as our method can more effectively utilize the text descriptions for tissue segmentation.

E. Effects of Different Category Descriptions for Tissue Segmentation

To assess how varying levels of semantic detail in textual descriptions influence tissue segmentation, we compare three types of category descriptions (1) Class Label: only use tissue names (*e.g.*, tumor) as category descriptions (2) Brief Descriptions (Bri. Des.): generate the brief descriptions of different tissue categories with the prompt “Describe {} in the histopathology images.”, where the placeholder is replaced with the tissue type. (3) Comprehensive Description (Com. Des.): our method that can generate comprehensive descriptions for different tissue categories (introduced in Section III-C).

As can be seen from Fig. 5, the adoption of comprehensive descriptions consistently leads to the highest segmentation performance across all datasets and unseen category settings. Such consistent performance improvement indicates that richer semantic information like color, shape and arrangement is critical for guiding the alignment between visual features and semantic prototypes, and thus can enhance tissue segmentation for both seen and unseen categories.

F. Effects of Different Pathology Vision-Language Models for Tissue Segmentation on Different Datasets

Fig. 6 compares the performance of two pathology vision-language models, CONCH (used in our study) and PLIP [66], on tissue segmentation across TNBC, HPBC, and CATCH datasets under varying numbers of unseen categories. The results show that both models achieve similar performance across different metrics *i.e.*, pACC, mIoU and hIoU. Such observations suggest that both pathology vision-language models show competitive power for zero-shot tissue segmentation task. In other words, the gains from pathology-language alignment are largely retained regardless of the specific model choice, supporting the robustness and general applicability of



Fig. 4. Visualization of the segmentation results on three chosen datasets. The white and red tags represent seen and unseen categories, respectively.

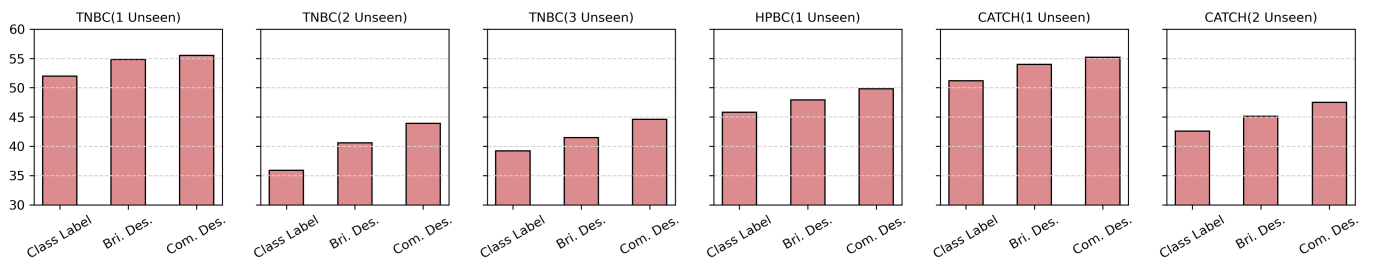


Fig. 5. Effects of different category descriptions for tissue segmentation. Bri. Des.: Brief description, Com. Des.: Comprehensive description.

our CONCH-based framework for zero-shot tissue segmentation task.

G. Specializations of Different Local Experts in MoLE Module

In the MoLE module of the proposed ZSPMLG, we design four different local experts aims at concentrating in different

tissue structures or categories of pathological images for tissue segmentation. To further validate the specialization of these four local experts, we exhibit the high-resolution patches with the largest weights (*i.e.*, $G_{local}(F_l^h)$) that different experts focus on in Fig. 7.

As can be seen from Fig. 7, different local experts are indeed specialized in capturing distinct structural patterns or semantic

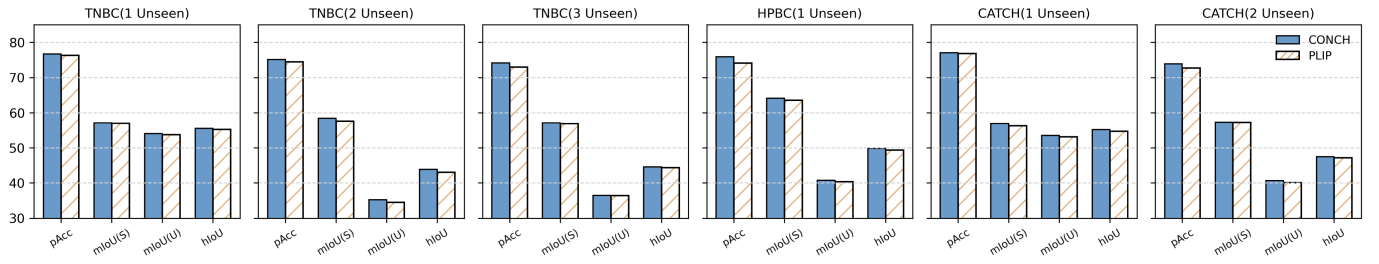


Fig. 6. Comparisons of applying different pathology vision-language models for tissue segmentation on different datasets.

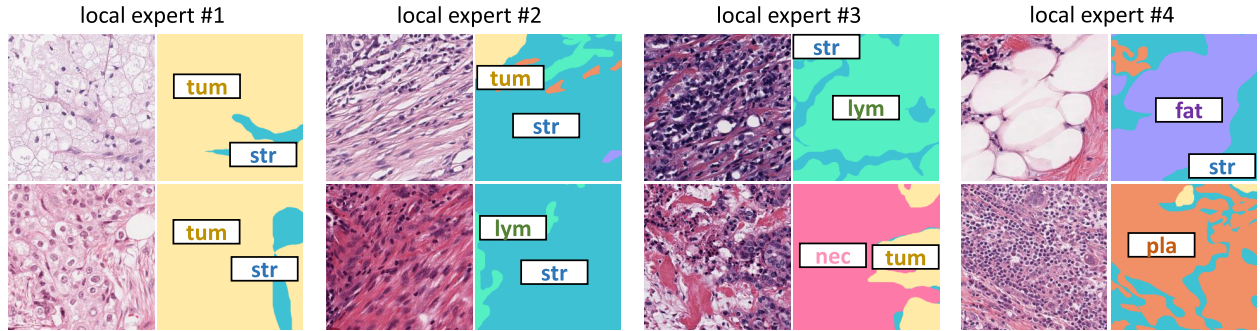


Fig. 7. Specializations of different local experts for the high magnification pathological images on TNBC dataset.

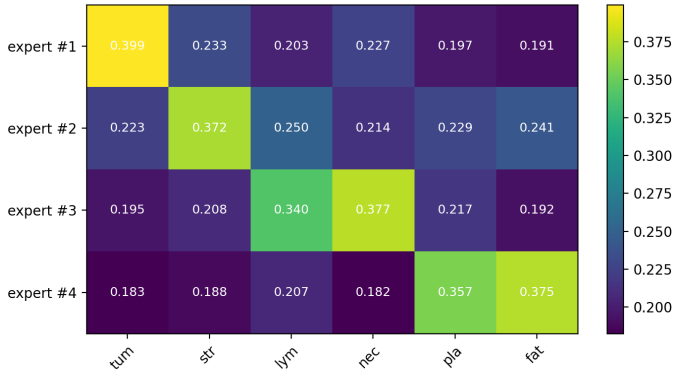


Fig. 8. Average routing weights for MoLE experts across different tissue categories on TNBC dataset.

regions in pathological images. For instance, Expert 1 tends to focus on tumor regions with epithelial arrangements, while Expert 2 is more responsive to stromal tissues with spindle-shaped cells. Moreover, it is obvious that Expert 3 specializes in identifying small and scattered regions such as lymphocytes clusters or necrotic areas that are often with dark and round nuclei. Finally, Expert 4 consistently emphasizes regions that are associated with unseen tissue categories (*i.e.*, fat and plasma). Such observations are consistent with the quantitative routing weights analysis presented in Fig. 8, where each expert exhibits a significantly higher activation probability for its preferred tissue type compared to others.

H. Effects of the Number of Experts in MoLE Module

As introduced in Section III-D, we set the number of experts in MoLE modules as 4. To evaluate the effects of expert

number in MoLE, we tune it from 1 to 5 with the interval of 1 and report the tissue segmentation results across different datasets under the setting of 1 unseen category. As can be observed from Fig. 9(a), the mean Intersection-over-Union (mIoU) for both seen and unseen categories consistently improves as the number of experts increases from 1 to 3. This suggests that the incorporation of a greater diversity of local experts enhances the model's ability to capture heterogeneous tissue structures. However, the segmentation results will be stable if the expert number is larger than 3. Such trend implies that the MoLE module can model tissue structure with a moderate number of local experts, and further increasing the local expert number will increase the computational complexity while less effective in improving the tissue segmentation performance.

I. Effects of Different Experts in MoGE Module

As mentioned in Section III-D, we design four experts in the MoGE module, where each expert corresponds to a way for fusing multi-scale pixel representations. To assess the contribution of each individual expert, we conduct a leave-one-out ablation study by iteratively removing each expert from the MoGE module and evaluating the segmentation performance. As can be seen from Fig. 9(b), the removal of any single expert will lead to the decline for tissue segmentation on both seen and unseen tissue categories, suggesting that each expert plays a vital and non-redundant role in the overall fusion process. These findings highlight the importance of expert diversity in the MoGE module, where different fusion modes collectively enhance the expressiveness and generalization capability of the global representation for tissue segmentation. We have also

TABLE III
ABLATION STUDY FOR THE PROPOSED MoLE AND MoGE MODULES ON TNBC DATASET

MoLE	MoGE	1 unseen				2 unseen				3 unseen			
		pAcc	mIoU(S)	mIoU(U)	hIoU	pAcc	mIoU(S)	mIoU(U)	hIoU	pAcc	mIoU(S)	mIoU(U)	hIoU
✗	✗	73.1	53.0	47.2	50.0	72.2	55.4	29.6	38.6	71.0	56.1	30.5	39.5
✓	✗	75.3	55.9	48.6	52.0	73.7	56.0	33.1	41.6	72.8	56.4	34.7	43.0
✗	✓	75.8	54.6	51.3	52.9	74.3	56.5	33.7	42.2	73.2	56.2	33.9	42.3
✓	✓	76.7	57.1	54.0	55.5	75.1	58.4	36.4	44.8	74.2	57.1	35.2	43.6

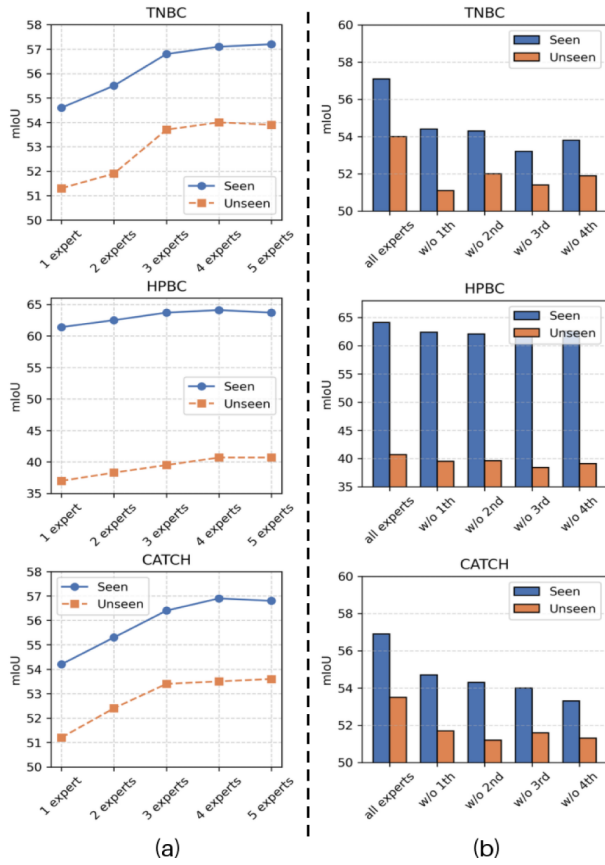


Fig. 9. Effects of experts in MoLE and MoGE under the settings of 1 unseen category. (a) The effects of different expert number in MoLE module. (b) The effects of each expert in MoGE module, where drop-“X” refers to removing expert “X” for model training.

TABLE IV
AVERAGE ROUTING PROBABILITY OF MoGE EXPERTS
ON TNBC DATASET

Expert #1	Expert #2	Expert #3	Expert #4
0.263	0.247	0.222	0.268

conducted a quantitative analysis by calculating the average selection frequency of each global expert, where each expert corresponds to a specific fusion strategy. As presented in Table IV, the global gating network does not converge to a

single dominant fusion strategy. Instead, it leverages different experts for different input patterns, indicating that the MoGE module maintains diverse fusion pathways to accommodate varying multi-scale characteristics.

J. Ablation Studies for the MoLE and MoGE Modules

In this section, we also discuss the effects of MoLE and MoGE modules on the TNBC dataset under different unseen category settings. The experimental results shown in Table III clearly verify that both of these two modules contribute positively to the zero-shot tissue segmentation tasks. Specifically, we can witness a significant improvement under the indexes of mIoU(S) and mIoU(U) if the MoLE module is included, showing the advantage of the Mixture of Local Experts that can help capture diverse local tissue structure for tissue segmentation. Similar observations can be obtained if we further add the MoGE modules to more flexibly combine multi-scale pixel-level representations. All these results again validate the effectiveness of MoLE and MoGE modules on zero-shot tissue segmentation tasks.

K. Effects of Fusing Multi-Scale Representations for Tissue Segmentation

In this section, we further discuss the effects of fusing multi-scale representations for tissue Segmentation. As a matter of fact, the low-resolution branch (2x down-sample branch) will be removed if the MoGE module is not involved, and the experimental results shown in Table III verify the advantage of combining multi-scale information for tissue segmentation on both seen and unseen categories.

V. DISCUSSION

A. Effects of Different Vision-Language Models for Tissue Segmentation on TNBC Datasets

We also compare CONCH with two general vision-language models i.e., CLIP [19] and CoCa [67], and two different medical vision-language models that are not specific to pathology images, i.e., MedCLIP [68] and PubMedCLIP [69] for zero-shot tissue segmentation on TNBC dataset. As can be seen from Fig. 10(a), both CLIP and CoCa perform worse than CONCH for different unseen settings. Such observations indicate that general-domain vision-language models, despite their strong performance on natural images, struggle

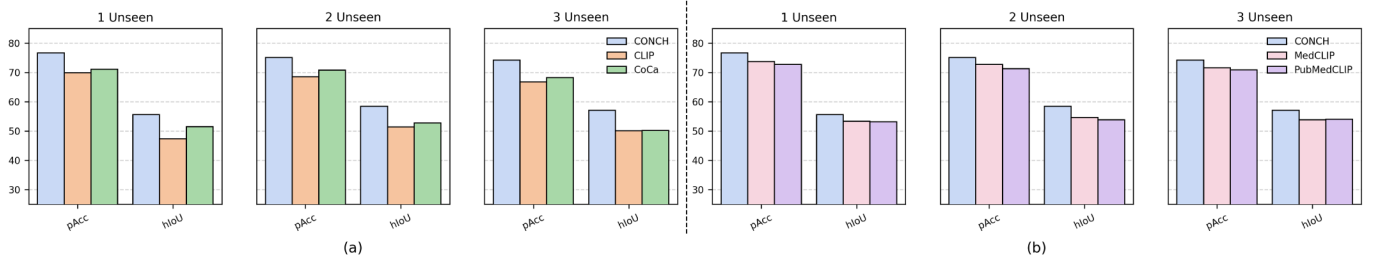


Fig. 10. Comparisons of different vision-language models (VLMs) for zero-shot tissue segmentation on TNBC dataset. (a) Comparisons of CONCH with general VLMs. (b) Comparison of CONCH with medical vision-language models that are not specific to pathology images.

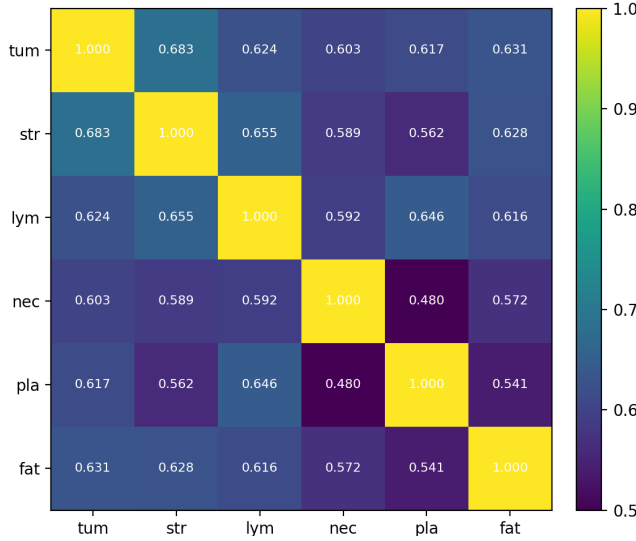


Fig. 11. Pairwise cosine similarities of different tissue descriptions using the pretrained CONCH model on TNBC dataset.

to generalize to the specialized domain of histopathology due to the significant domain gap in visual appearance and semantic concepts. Moreover, as shown in Fig. 10(b), both PubMedCLIP [69] and MedCLIP [68] also perform worse than CONCH, which further demonstrate the advantage of adopting pathology-specific foundation model for the task of zero-shot tissue segmentation.

B. Semantic Similarity Among Different Tissue Categories

To quantify the semantic relationship between seen and unseen classes, we compute pairwise cosine similarities of different tissue descriptions in TNBC dataset using pretrained CONCH model. The experimental results shown in Fig. 11 indicate that the similarity values between seen (*i.e.*, tum, str, lym) and unseen categories (*i.e.*, nec, pla and fat) are similar to those among seen categories and among unseen categories themselves, suggesting no systematic semantic bias in our split. This ensures that the difficulty of the zero-shot task arises from the inherent generalization challenge rather than from artificially inflated semantic gaps, providing a fair and realistic evaluation of model performance.

C. Effects of LLM-Generated and Handcrafted Descriptions for Tissue Segmentation

In this section, we have compared the use of LLM-generated descriptions against manually curated descriptions of different

tissue types (*i.e.*, ZSPMLG-M) derived from the book in [70]. As can be seen from Fig. 12(a), using LLM-generated descriptions achieves similar segmentation performance in comparison with the results relying on handcrafted features. Such results suggest that the LLM-generated descriptions exhibit high reliability for zero-shot tissue segmentation task, making them a trustworthy alternative to manual prompts.

D. Effects of Contradictory Textual Descriptions for Tissue Segmentation

To assess the robustness of our method against contradictory or misleading text description, we randomly permute the comprehensive descriptions of different category labels where the semantic prototypes are misaligned with their corresponding tissue types. As shown in Fig. 12(b), under such contradictory settings, ZSPMLG experiences significant performance degradation across different experimental settings on TNBC dataset. These results demonstrate the critical importance of accurate semantic-visual alignment for zero-shot tissue segmentation on pathological images.

E. Effects of ZSPMLG for Cross Dataset Zero-Shot Tissue Segmentation

In this section, we conduct extensive experiments to assess the generalization ability of ZSPMLG across different datasets. Here, our model is trained exclusively on the HPBC dataset containing three seen categories (*i.e.*, tumor, stroma, lymphocyte infiltrate) and tested on the TNBC dataset that contains both the seen categories in HPBC and additional three novel unseen categories (*i.e.*, necrosis, plasma cells, fat) that are not involved in HPBC. The experimental results presented in Table V show that our method still significantly outperforms the comparing approaches on both the seen and unseen categories. Such observations indicates that our model not only adapts better to new data distributions but also learns more fundamental, category-agnostic features that generalize to unseen classes, thereby confirming its strong cross-dataset robustness.

F. Effects of ZSPMLG for Cross Organ Zero-Shot Tissue Segmentation

In this section, we conduct a more challenging out-of-distribution experiment by training the model on the breast histopathology dataset HPBC, and evaluating it on the lung

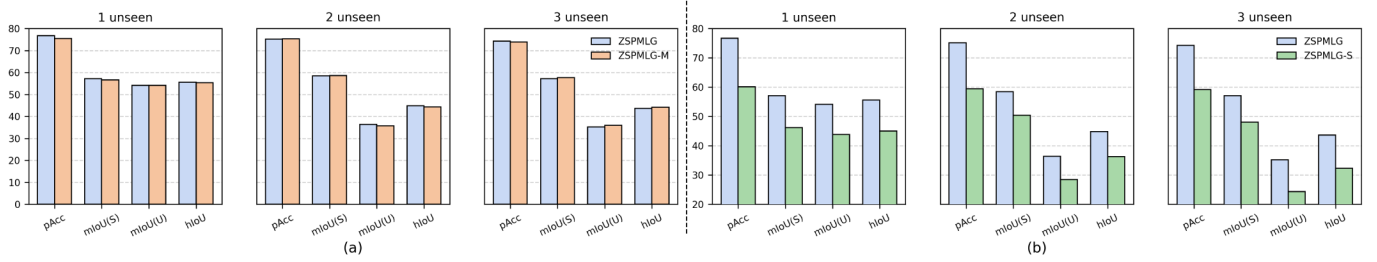


Fig. 12. (a) Comparison of using LLM-generated descriptions and handcrafted descriptions of different tissue types on TNBC dataset. (b) Effects of the contradictory descriptions of tissues for zero-shot tissue segmentation on TNBC dataset.

TABLE V
EFFECTS OF ZSPMLG FOR CROSS DATASET ZERO-SHOT
TISSUE SEGMENTATION

Method	pAcc	mIoU(S)	mIoU(U)	hIoU
USE [71]	67.2	50.9	29.6	37.4
TagCLIP [21]	68.5	51.2	30.5	38.2
ZSPMLG(ours)	70.7	53.4	32.0	40.0

TABLE VI
EFFECTS OF ZSPMLG FOR CROSS ORGAN
ZERO-SHOT TISSUE SEGMENTATION

Method	pAcc	mIoU(S)	mIoU(U)	hIoU
USE [71]	66.2	57.9	34.9	43.5
TagCLIP [21]	67.8	57.7	36.3	44.6
ZSPMLG(ours)	71.3	58.8	38.2	46.3

adenocarcinoma dataset *i.e.*, LUAD-HistoSeg [72], which introduces a significant organ shift from breast to lung tissue. In LUAD-HistoSeg dataset, we follow the split of seen/unseen categories in HPCB, where the seen categories include background, tumor and stroma, while lymphocyte infiltrate is treated as the unseen category. As shown in Table VI, our method maintains superior performance over the state-of-the-art zero-shot segmentation baselines including TagCLIP [21] and USE [71]. Notably, when comparing these cross-organ results to the same-organ results on HPCB reported in Table II, the performance drop is relatively modest given the substantial domain shift from breast to lung tissue. Such results indicate that our pathology-specific semantic prototypes and multi-scale expert design maintain strong generalization capabilities under challenging organ-level domain shifts, further validating the robustness and practical value of our approach for real-world pathology applications.

G. Sensitivity Analysis of the Prompts

Since the LLMs can produce different outputs across different runs or sessions, to evaluate the sensitivity of the prompts, we have used 5 different descriptions generated from the same instruction prompt as the input to the text encoder during inference, while keeping all other settings unchanged. The results are shown in Table VII.

TABLE VII
SENSITIVITY ANALYSIS OF THE PROMPTS ON TNBC DATASET

1 Unseen(hIoU)	2 Unseen(hIoU)	3 Unseen(hIoU)
54.63 $\pm$ 0.95	44.53 $\pm$ 0.93	42.87 $\pm$ 0.64

TABLE VIII
COMPARISON OF ZSPMLG WITH ITS VARIANT ZSPMLG-N BY ADDING
GAUSSIAN NOISE (MEAN = 0, STD = 0.01) ON THE GENERATED
PROTOTYPES FOR TNBC DATASET

Dataset	Metric	ZSPMLG	ZSPMLG-N
1 Unseen	pAcc	76.7	75.9
	hIoU	55.6	55.0
2 Unseen	pAcc	75.1	73.8
	hIoU	44.8	43.9
3 Unseen	pAcc	74.2	73.7
	hIoU	43.6	42.9

As can be seen from Table VII, the segmentation performance is only slightly affected by different generated descriptions with the same prompt instruction, suggesting that our method is robust to the generated description when the prompt instruction is fixed.

H. Robustness Analysis of the Generated Prototype

In this section, we compare our proposed ZSPMLG against a perturbed baseline, ZSPMLG-N, which introduces Gaussian noise ($\mu = 0, \sigma = 0.01$) to the VLM-generated prototypes. As can be seen in Table VIII, ZSPMLG-N achieves comparable performance to the original ZSPMLG, suggesting that our ZSPMLG is robust to the variance of prototypes.

I. Effects of Prototype Dimensionality for Zero-Shot Tissue Segmentation

To justify the choice of the 64-dimensional prototype for the zero-shot tissue segmentation task, we conduct additional experiments by adding an extra convolution layer to the MoLE module that can adjust the dimension of the prototypes to 32 and 128. As shown in Fig. 13, the 64-dimensional prototype consistently achieves the best overall performance

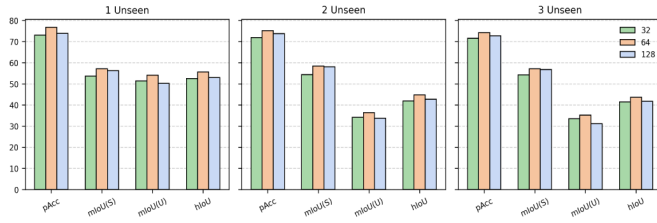


Fig. 13. Impact of prototype dimensionality for zero-shot tissue segmentation on TNBC dataset.

TABLE IX
COMPARISON OF ZSPMLG UNDER ZERO-SHOT AND FULLY SUPERVISED SETTINGS ON TNBC DATASET

Settings	Zero-Shot		Fully Supervised	
	pAcc	hIoU	pAcc	hIoU
1 Unseen	76.7	55.6	81.5	59.7
2 Unseen	75.1	44.8	80.4	50.3
3 Unseen	74.2	43.6	80.9	49.4

across all experimental settings, while higher/lower prototype dimensions lead to slight but consistent performance degradation. These results verify the effectiveness of choosing a 64-dimensional prototype space for discriminative feature learning on zero-shot tissue segmentation task.

J. Comparison With Fully Supervised Settings

In this section, we compare our method under zero-shot and fully supervised (FS) settings. In FS settings, all tissue categories are annotated with ground-truth labels and the model is trained on the full set of labeled data without any unseen category constraints. As can be seen from Table IX, the fully supervised (FS) setting consistently yields moderately higher performance across all metrics compared to the zero-shot setting (ZSS), which is expected given that fully supervised methods have access to ground-truth annotations for all classes during training. Nevertheless, the segmentation performance of zero-shot setting is still competitive with FS. For instance, for the setting of 1 unseen category, the hIoU difference between ZSS and FS is only 4.1%.

K. Comparison With the Existing Studies Under Inductive Settings

We additionally compare our method with its competitors under a strict inductive setting, where textual descriptions of unseen categories are excluded from training and introduced only at inference stage. The results are listed in Table X.

As can be seen from Table X, although the segmentation performance under inductive settings drops in comparison with the transductive settings, our method still outperforms the baselines. This additional experiment confirms that the superiority of our approach is not merely due to the transductive access to unseen-class text descriptions, but rather stems from the intrinsic effectiveness of the proposed model design.

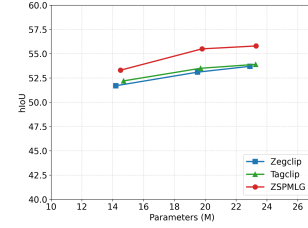


Fig. 14. Comparison of the scaling curves among ZegCLIP, TagCLIP, and our ZSPMLG.

L. Comparison of Computational Complexity Among Different Methods

In this section, we compare the computational complexity and segmentation performance between our method and the existing studies. As shown in Table XI, while our proposed ZSPMLG indeed requires moderately more parameters (19.8M vs. 13.8M/15.2M) and incurs moderately higher latency (43.6ms vs. 31.4ms/32.4ms) compared to the state-of-the-art comparing methods, *i.e.*, ZegCLIP and TagCLIP, this modest trade-off in computational cost brings substantial gains in segmentation performance *i.e.*, ZSPMLG achieves 55.5 hIoU, outperforming ZegCLIP (51.7) and TagCLIP (52.2) by a significant margin of 3.3 to 3.8 percentage points. We believe this performance improvement justifies the moderate higher resource consumption, as the primary objective of our work is to advance the state-of-the-art in zero-shot tissue segmentation. Moreover, to address whether our method maintains its advantage when competing methods are scaled to a comparable parameter count, we have conducted a new scaling experiment across three parameter levels (14M, 20M, and 23M) for ZegCLIP, TagCLIP, and our method. As shown in Fig. 14, our method consistently achieves the highest mIoU at different parameter levels. These results further confirm that our method not only achieves superior performance at a given parameter budget but also exhibits a more favorable scaling behavior, with performance improving steadily over different parameter capacities.

M. Effects of Multi-Scale Expert Mechanism for the Segmentation of Different Tissue Categories

In this section, we discuss the effects of multi-scale expert mechanism for the segmentation of different tissue categories. By comparing the “w/o MoGE” and “w/o MoLE” settings against the full ZSPMLG model, we analyze how the removal of each expert module individually impacts the segmentation accuracy across six distinct tissue types. As shown in Table XII, lymphocytic infiltrate (lym.) and plasma cells (pla.) exhibit the largest performance gains when both MoLE and MoGE modules are incorporated, with mIoU improvements of + 3.8% (from 42.4% to 46.2%) and + 5.4% (from 29.7% to 35.1%), respectively. In contrast, categories such as fat (fat.) and stroma (str.), which are predominantly characterized by consistent morphological features recognizable at a single scale, show comparatively modest gains. This pattern aligns with histopathology practice where lymphocytes and plasma cells are with more complex structure requiring

TABLE X

COMPARISON OF ZSPMLG WITH THE EXISTING ZERO-SHOT SEGMENTATION METHODS UNDER INDUCTIVE SETTINGS ON TNBC DATASET. w2v: WORD2VEC. mIoU(S): mIoU ON SEEN CATEGORIES. mIoU(U): mIoU ON UNSEEN CATEGORIES. hIoU: HARMONIC mIoU

Method	Text Encoder	TNBC											
		1 unseen				2 unseen				3 unseen			
		pAcc	mIoU(S)	mIoU(U)	hIoU	pAcc	mIoU(S)	mIoU(U)	hIoU	pAcc	mIoU(S)	mIoU(U)	hIoU
SPNet [61]	w2v	57.4	35.2	26.9	30.5	58.3	41.5	18.2	25.3	56.5	42.9	16.7	24.0
ZS3 [41]	w2v	59.9	34.1	30.2	32.0	58.8	40.3	24.3	30.3	56.3	40.3	20.2	26.9
CaGNet [62]	w2v	63.3	39.2	34.1	36.5	60.6	45.2	24.6	31.9	60.2	46.9	23.4	31.2
SaLiP [49]	CONCH	70.0	42.7	41.8	42.2	68.9	51.2	25.9	34.4	69.7	53.5	26.4	35.4
ZegCLIP [20]	CONCH	74.6	50.6	48.9	49.7	73.2	55.9	29.5	38.6	72.4	55.6	28.3	37.5
USE [71]	CONCH	74.5	51.8	49.3	50.5	74.2	57.3	28.1	37.7	72.6	56.0	28.6	37.9
TagCLIP [21]	CONCH	74.8	53.1	50.4	51.7	73.5	57.4	30.1	39.5	72.8	56.2	28.9	38.2
ZSPMLG(ours)	CONCH	75.8	55.9	51.2	53.4	73.7	57.8	33.3	42.3	72.3	55.4	30.6	39.4

TABLE XI

COMPARISON OF MODEL COMPLEXITY AND PERFORMANCE ON TNBC DATASET

Model	Params	FLOPs	Latency	Peak mem	hIoU
ZegCLIP [20]	13.8M	110.4G	31.4ms	768.2M	51.7
TagCLIP [21]	15.2M	112.4G	32.4ms	760.2M	52.2
ZSPMLG(ours)	19.8M	120.3G	43.6ms	870.1M	55.5

TABLE XII

EFFECTS OF MULTI-SCALE EXPERT MECHANISM FOR THE SEGMENTATION OF DIFFERENT TISSUE CATEGORIES ON TNBC DATASET

Settings	Tum.	Str.	Lym.	Nec.	Pla.	Fat
w/o MoGE	70.1	50.9	42.9	60.1	29.7	36.5
w/o MoLE	70.8	51.2	42.4	61.7	31.2	36.1
ZSPMLG	73.1	51.8	46.2	62.3	35.1	37.6

low-magnification assessment for spatial distribution and high-magnification verification for cytomorphological details [73], [74], whereas fat and stromal regions are typically identifiable at a fixed magnification. Thus, the multi-scale expert mechanism is most advantageous for complex tissue categories with scale-dependent visual characteristics that require cross-scale integration. It is worth noting that the largest performance gains from MoLE and MoGE occur for unseen categories plasma cells. Such substantial improvements can also demonstrate that both MoLE and MoGE modules possess strong generalization on unseen categories.

N. Limitations

While the proposed ZSPMLG demonstrates good zero-shot tissue segmentation performance across multiple datasets, the limitations or failure modes of our method lie in the following two aspects. Firstly, as shown in Table I and Table II, the hIoU values consistently decline as the number of unseen categories increases, due to the corresponding reduction in seen categories available for supervision. Since pixel-level annotations are only accessible for seen categories during

training, fewer seen classes limit the model's ability to learn robust semantic-visual correspondences necessary for generalizing to unseen categories. Notably, this limitation is not unique to ZSPMLG but reflects a broader challenge inherent to zero-shot segmentation task [20]. Secondly, the current framework faces challenges when applied across substantially different pathology modalities, such as from H&E to IHC staining. This is because the existing pathology vision-language models (*e.g.*, CONCH [28]) are trained primarily on H&E-stained images, our model similarly relies on learning visual-semantic alignment within the H&E modality using these pretrained encoders and text-guided prototypes. When the staining modality changes to IHC, both image appearance and the corresponding semantic descriptions differ significantly from the H&E-based pretraining distribution, making the learned knowledge much less reusable. In future, we plan to fine-tune the H/E stained foundation model with IHC stained images to make it applicable on IHC-stained pathological image segmentation tasks.

VI. CONCLUSION

In this study, we propose ZSPMLG, a novel vision-language foundation model-based zero-shot tissue segmentation framework that effectively addresses the challenges of segmenting unseen tissue categories in pathological images. By considering that the textual descriptions of specific tissue categories can be associated with visual patterns observable in pathology images at different magnification scales, we learn and combine the pixel-level representations of different scales by the mixture of MoLE and MoGE modules, where MoLE performs the specialized decoding that can map individual scale patch-level representations to dense pixel-level representation, while MoGE aims at fusing the multi-scale representations together. The experimental results on all three datasets verify the advantage of our method for segmenting both seen and unseen tissue categories in comparison with the existing studies.

REFERENCES

- [1] F. Xing and L. Yang, "Robust nucleus/cell detection and segmentation in digital pathology and microscopy images: A comprehensive review," *IEEE Rev. Biomed. Eng.*, vol. 9, pp. 234–263, 2016.

- [2] R. Xu et al., "Histopathological tissue segmentation of lung cancer with bilinear CNN and soft attention," *BioMed Res. Int.*, vol. 2022, no. 1, 2022, Art. no. 7966553.
- [3] L. Chan, M. Hosseini, C. Rowsell, K. Plataniotis, and S. Damaskinos, "HistoSegNet: Semantic segmentation of histological tissue type in whole slide images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 10661–10670.
- [4] Q. D. Vu et al., "Methods for segmentation and classification of digital microscopy tissue images," *Frontiers Bioengineering Biotechnol.*, vol. 7, p. 53, Apr. 2019.
- [5] J. Fan et al., "DIPathMamba: A domain-incremental weakly supervised state space model for pathology image segmentation," *Med. Image Anal.*, vol. 103, Jul. 2025, Art. no. 103563.
- [6] S. Wang, D. M. Yang, R. Rong, X. Zhan, and G. Xiao, "Pathology image analysis using segmentation deep learning algorithms," *Amer. J. Pathol.*, vol. 189, no. 9, pp. 1686–1698, 2019.
- [7] K. Jagadeesh and A. Rajendran, "Improved model for genetic algorithm-based accurate lung cancer segmentation and classification," *Comput. Syst. Sci. Eng.*, vol. 45, no. 2, pp. 2017–2032, 2023.
- [8] Y. Wu et al., "Recent advances of deep learning for computational histopathology: Principles and applications," *Cancers*, vol. 14, no. 5, p. 1199, Feb. 2022.
- [9] W. Shao et al., "Integrative analysis of pathological images and multi-dimensional genomic data for early-stage cancer prognosis," *IEEE Trans. Med. Imag.*, vol. 39, no. 1, pp. 99–110, Jan. 2020.
- [10] D. Tacon, "How I peer into the geometry behind computer vision," *Nature*, vol. 634, no. 8035, p. 2, 2024.
- [11] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [12] Y. Luo et al., "Deep learning network for NMR spectra reconstruction in time-frequency domain and quality assessment," *Nature Commun.*, vol. 16, no. 1, p. 2342, Mar. 2025.
- [13] A. Voulodimos, N. Doulamis, A. Doulamis, and E. Protopapadakis, "Deep learning for computer vision: A brief review," *Comput. Intell. Neurosci.*, vol. 2018, Oct. 2018, Art. no. 7068349.
- [14] A. Paproki, O. Salvado, and C. Fookes, "Synthetic data for deep learning in computer vision & medical imaging: A means to reduce data bias," *ACM Comput. Surv.*, vol. 56, no. 11, pp. 1–37, Nov. 2024.
- [15] X. Liu et al., "MSMTSeg: Multi-stained multi-tissue segmentation of kidney histology images via generative self-supervised meta-learning framework," *IEEE J. Biomed. Health Informat.*, vol. 29, no. 6, pp. 3906–3917, Jun. 2024.
- [16] M. van Rijthoven, M. Balkenhol, K. Siliņa, J. van der Laak, and F. Ciompi, "HookNet: Multi-resolution convolutional neural networks for semantic segmentation in histopathology whole-slide images," *Med. Image Anal.*, vol. 68, Feb. 2021, Art. no. 101890.
- [17] V. Thorsson et al., "The immune landscape of cancer," *Immunity*, vol. 48, no. 4, pp. 812–830, 2018.
- [18] J. Y. Cheng, J. T. Abel, U. G. Balis, D. S. McClintock, and L. Pantanowitz, "Challenges in the development, deployment, and regulation of artificial intelligence in anatomic pathology," *Amer. J. Pathol.*, vol. 191, no. 10, pp. 1684–1692, 2021.
- [19] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.
- [20] Z. Zhou, Y. Lei, B. Zhang, L. Liu, and Y. Liu, "ZegCLIP: Towards adapting CLIP for zero-shot semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11175–11185.
- [21] J. Li, P. Chen, S. Qian, S. Liu, and J. Jia, "TagCLIP: Improving discrimination ability of zero-shot semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 11287–11297, Dec. 2024.
- [22] J. Ding, N. Xue, G.-S. Xia, and D. Dai, "Decoupling zero-shot semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Sep. 2022, pp. 11583–11592.
- [23] C. Zhou, C. C. Loy, and B. Dai, "Extract free dense labels from clip," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 696–712.
- [24] H. Ding et al., "MeViS: A multi-modal dataset for referring motion expression video segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 12, pp. 11400–11416, Dec. 2025.
- [25] H. Ding et al., "MOSEv2: A more challenging dataset for video object segmentation in complex scenes," 2025, *arXiv:2508.05630*.
- [26] W. Cong, Y. Cong, J. Dong, and G. Sun, "Towards efficient and effective interactive 3D segmentation," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2026, vol. 40, no. 5, pp. 3425–3433.
- [27] W. Cong, Y. Cong, and Y. Ren, "Lightweight class incremental semantic segmentation without catastrophic forgetting," *IEEE Trans. Image Process.*, vol. 34, pp. 4566–4579, 2025.
- [28] M. Y. Lu et al., "A visual-language foundation model for computational pathology," *Nature Med.*, vol. 30, no. 3, pp. 863–874, Mar. 2024.
- [29] M. Schuveling et al., "A novel dataset for nuclei and tissue segmentation in melanoma with baseline nuclei segmentation and tissue segmentation benchmarks," *GigaScience*, vol. 14, Jan. 2025, Art. no. g1af011.
- [30] T. J. Fuchs and J. M. Buhmann, "Computational pathology: Challenges and promises for tissue analysis," *Computerized Med. Imag. Graph.*, vol. 35, nos. 7–8, pp. 515–530, 2011.
- [31] Z. Lu et al., "BrcaSeg: A deep learning approach for tissue quantification and genomic correlations of histopathological images," *Genomics, Proteomics Bioinf.*, vol. 19, no. 6, pp. 1032–1042, 2021.
- [32] H. Chen et al., "DCAN: Deep contour-aware networks for object instance segmentation from histology images," *Med. Image Anal.*, vol. 36, pp. 135–146, Sep. 2017.
- [33] J. Saltz et al., "Spatial organization and molecular correlation of tumor-infiltrating lymphocytes using deep learning images," *Cell Rep.*, vol. 23, no. 1, pp. 181–193, 2018.
- [34] R. Schmitz et al., "Multi-scale fully convolutional neural networks for histopathology image segmentation: From nuclear aberrations to the global tissue architecture," *Med. Image Anal.*, vol. 70, Aug. 2021, Art. no. 101996.
- [35] Q. D. Vu and J. T. Kwak, "A dense multi-path decoder for tissue segmentation in histopathology images," *Comput. Methods Programs Biomed.*, vol. 173, pp. 119–129, Aug. 2019.
- [36] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," in *Proc. Int. MICCAI Brainlesion Workshop*, 2021, pp. 272–284.
- [37] X. Hu et al., "Expert knowledge-guided multi-granularity multi-scale fusion for weakly-supervised histological segmentation," *Inf. Fusion*, vol. 125, Jan. 2026, Art. no. 103432.
- [38] X. Gu, L. Wang, Z. Deng, Y. Cao, X. Huang, and Y.-M. Zhu, "Adaptive spatial and frequency experts fusion network for medical image fusion," *Biomed. Signal Process. Control*, vol. 96, Oct. 2024, Art. no. 106478.
- [39] X. Guan et al., "Global-local consistent semi-supervised segmentation of histopathological image with different perturbations," *Pattern Recognit.*, vol. 155, Jun. 2024, Art. no. 110696.
- [40] Y. Kang et al., "Exploring unbiased activation maps for weakly supervised tissue segmentation of histopathological images," *IEEE Trans. Med. Imag.*, vol. 44, no. 6, pp. 2631–2642, Jun. 2025.
- [41] M. Bucher, T.-H. Vu, M. Cord, and P. Pérez, "Zero-shot semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–14.
- [42] W. Ren, Y. Tang, Q. Sun, C. Zhao, and Q.-L. Han, "Visual semantic segmentation based on few/zero-shot learning: An overview," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 5, pp. 1106–1126, May 2024.
- [43] W. Wang, V. W. Zheng, H. Yu, and C. Miao, "A survey of zero-shot learning: Settings, methods, and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 1–37, Mar. 2019.
- [44] L. Zhu et al., "WeakCLIP: Adapting CLIP for weakly-supervised semantic segmentation," *Int. J. Comput. Vis.*, vol. 133, no. 3, pp. 1085–1105, Mar. 2025.
- [45] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.
- [46] Z. Zhang, Y. Zhou, J. Yue, R. Zhang, and J. Ma, "IIIM-SAM: Zero-shot texture anomaly detection without external prompts," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 14610–14622, 2025.
- [47] T. Koleilat, H. Asgariandehkordi, H. Rivaz, and Y. Xiao, "MedCLIP-sam: Bridging text and image towards universal medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*, 2024, pp. 643–653.
- [48] B. Towle, X. Chen, and K. Zhou, "SimSAM: Zero-shot medical image segmentation via simulated interaction," in *Proc. IEEE Int. Symp. Biomed. Imag. (ISBI)*, May 2024, pp. 1–5.
- [49] S. Aleem et al., "Test-time adaptation with SaLIP: A cascade of SAM and CLIP for zero-shot medical image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 5184–5193.
- [50] A. Hamrani and A. Godavarty, "Self-attention diffusion models for zero-shot biomedical image segmentation: Unlocking new frontiers in medical imaging," *Bioengineering*, vol. 12, no. 10, p. 1036, Sep. 2025.
- [51] D. Dai et al., "PathologyVlm: A large vision-language model for pathology image understanding," *Artif. Intell. Rev.*, vol. 58, no. 6, p. 186, Mar. 2025.

- [52] R. J. Chen et al., "Towards a general-purpose foundation model for computational pathology," *Nature Med.*, vol. 30, no. 3, p. 37, Mar. 2024.
- [53] J. Wu et al., "Learning heterogeneous tissues with mixture of experts for gigapixel whole slide images," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 5144–5153.
- [54] B. Fu et al., "Multimodal dynamic fusion framework for survival prediction in clear cell renal cell carcinoma," *Inf. Fusion*, vol. 132, Aug. 2026, Art. no. 104217.
- [55] X. Zhou et al., "Knowledge-enhanced pretraining for vision-language pathology foundation model on cancer diagnosis," *Cancer Cell*, vol. 44, no. 4, pp. 777–791, Apr. 2026.
- [56] J. Tang, W. Shao, and D. Zhang, "Cancer survival analysis via zero-shot tumor microenvironment segmentation on low-resolution whole slide pathology images," in *Proc. 39th Annu. Conf. Neural Inf. Process. Syst.*, 2025, pp. 1–12.
- [57] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.
- [58] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *J. Mach. Learn. Res.*, vol. 23, no. 120, pp. 1–39, 2022.
- [59] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
- [60] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
- [61] Y. Xian, S. Choudhury, Y. He, B. Schiele, and Z. Akata, "Semantic projection network for zero{-} and few-label semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8248–8257.
- [62] Z. Gu, S. Zhou, L. Niu, Z. Zhao, and L. Zhang, "Context-aware feature generation for zero-shot semantic segmentation," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 1921–1929.
- [63] M. Amgad et al., "Structured crowdsourcing enables convolutional segmentation of histology images," *Bioinformatics*, vol. 35, no. 18, pp. 3461–3467, Sep. 2019.
- [64] F. Wilm et al., "Multi-scanner canine cutaneous squamous cell carcinoma histopathology dataset," in *Informatik Aktuell*, 2023, pp. 206–211.
- [65] Y. Xian, C. H. Lampert, B. Schiele, and Z. Akata, "Zero-shot learning—A comprehensive evaluation of the good, the bad and the ugly," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 9, pp. 2251–2265, Sep. 2019.
- [66] Z. Huang, F. Bianchi, M. Yuksekgonul, T. J. Montine, and J. Zou, "A visual-language foundation model for pathology image analysis using medical Twitter," *Nature Med.*, vol. 29, no. 9, pp. 2307–2316, Sep. 2023.
- [67] J. Yu, Z. Wang, V. Vasudevan, L. Yeung, M. Seyedhosseini, and Y. Wu, "CoCa: Contrastive captioners are image-text foundation models," 2022, *arXiv:2205.01917*.
- [68] Z. Wang, Z. Wu, D. Agarwal, and J. Sun, "MedCLIP: Contrastive learning from unpaired medical images and text," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2022, pp. 3876–3887.
- [69] S. Eslami, C. Meinel, and G. De Melo, "PubMedCLIP: How much does clip benefit visual question answering in the medical domain?," in *Proc. Findings Assoc. Comput. Linguistics, EACL*, 2023, pp. 1151–1163.
- [70] V. Kumar, A. K. Abbas, N. Fausto, and J. C. Aster, *Robbins and Cotran Pathologic Basis of Disease, Professional Edition E-Book*. Amsterdam, The Netherlands: Elsevier, 2014.
- [71] X. Wang et al., "USE: Universal segment embeddings for open-vocabulary image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 4187–4196.
- [72] S. Wazir and M. M. Fraz, "HistoSeg: Quick attention with multi-loss function for multi-structure segmentation in digital histology images," in *Proc. 12th Int. Conf. Pattern Recognit. Syst. (ICPRS)*, Jun. 2022, pp. 1–7.
- [73] F. B. Smith and A. Elnawawi, "A counting strategy for estimating plasma cell number in cd138-stained bone marrow core biopsy sections," *Ann. Clin. Lab. Sci.*, vol. 38, no. 2, pp. 138–142, 2008.
- [74] K. O. Alsaad and D. Ghazarian, "My approach to superficial inflammatory dermatoses," *J. Clin. Pathol.*, vol. 58, no. 12, pp. 1233–1241, Dec. 2005.